Carl Friedrich Gauß — Werke, Band I (Alternate) Nachlass: Conforme Abbildung des Sphäroids in der Ebene

Sections [3.]–[13.], expanded from the bracketed-gloss summary

Expanded against the original scan
(`carlfriedrichga01gausgoog9.pdf`, pp. 144–158)

This document expands the editorial `[bracketed glosses]` that the earlier typeset draft of `gauss_b01a_ch4.tex` (and the closing paragraphs of `ch3.tex`) had used to summarise multi-page derivations. The expansions here are typeset directly from the Hannover/Göttingen academy print of the Nachlass paper *Conforme Abbildung des Sphäroids in der Ebene* (Werke Band I Alternate, pages 144–158 of the scan). The prior `[bracketed glosses]` are kept as marginal cues so the reader can locate the corresponding stub in the source TeX; everything that was formerly inside or summarised by a gloss is now in upright running text with the displayed equations restored.

Setting and notation (carried in from S. 144).

Throughout the developments that follow, the conventions of Gauss on page 144 of the Nachlass are kept verbatim. To condense the writing one sets, in agreement with the section on the geodätische Linie,

$$\frac{\sqrt{(1-ee\sin\varphi^2)}}{a\cos\varphi}=t,\qquad \sin\varphi=s,$$

and writes the higher derivatives of the conform-map function f in the form

$$f^n(x)=t^n\,u,$$

where the auxiliary u is the rational function of s tabulated on page 144 of the scan (and reproduced in the source TeX, sect. [1.]). The recursion

$$f^{n+1}(x)=t^{n+1}\Big\{nus+\frac{\partial u}{\partial s}\cdot\frac{(1-ss)(1-ee\,ss)}{1-ee}\Big\}$$

generates each successive derivative, with the abbreviation $f^n(x)=tnu$ giving the table

n	u
1	1
2	s
3	$1+ss+ee(1-ss)^2$
4	$5s+s^3+ee(1-ss)^2s$

This is the table that the bracketed gloss `[Das Gesetz der Differentiation ist das folgende.]` in the source TeX abbreviates; it is written out here so that the formulae of sects. [6.]–[10.] below remain self-contained.

[6.] Berechnung des Vergrösserungsverhältnisses n (scan pp. 152–153, was a five-line gloss in the source TeX)

Editorial gloss in the source: `[Berechnung des Vergroesserungsverhaeltnisses n.]`; `[n ist das Verhaeltniss eines Linearelements in der Ebene zu dem entsprechenden Element auf dem Ellipsoid.]`

The quantity n is the ratio of a linear element in the plane to the corresponding element on the ellipsoid. With the abbreviations

$$f'(x) = a, \quad f''(x) = b, \quad f'''(x) = c, \quad f^{\mathrm{IV}}(x) = d, \quad \text{etc.},$$

one sets

$$\xi = x - w,$$

and from the conformality requirement $f(x+iy) = f(x-w)+i\lambda$ one obtains, by separating real and imaginary parts,

$$f(x) - \tfrac{1}{2}byy + \tfrac{1}{24}dy^4 - \tfrac{1}{720}fy^6 \ldots = f(x) - aw + \tfrac{1}{2}bww - \tfrac{1}{6}cw^3 \ldots$$

Setting

$$at = \tfrac{1}{2}byy - \tfrac{1}{24}dy^4 + \tfrac{1}{720}fy^6 \ldots,$$

one has

$$t = w - \frac{b}{2a}ww + \frac{c}{6a}w^3 \ldots$$

or, by inversion,

$$w = t + \frac{b}{2a}tt + \Big(\frac{bb}{2aa} - \frac{c}{6a}\Big)t^3 \ldots;$$

whence

$$w = \frac{b}{2a}yy + \Big(\frac{b^3}{8a^3} - \frac{d}{24a}\Big)y^4 + \Big(\frac{b^5}{16a^5} - \frac{b^2c}{48a^4} - \frac{bbd}{48a^3} + \frac{f}{720a}\Big)y^6 \ldots$$

and

$$\xi = x - \frac{b}{2a}yy + \frac{aad - 3b^3}{24a^3}y^4 - \frac{a^2f - 15aabbd - 15ab^3c + 45b^5}{720a^5}y^6 \ldots.$$

Since

$$f'(\xi) = f'(x) - wf''(x) + \tfrac{1}{2}wwf'''(x) - \tfrac{1}{6}w^3f^{\mathrm{IV}}(x) + \cdots = a - bw + \tfrac{1}{2}cww - \tfrac{1}{6}dw^3 + \cdots,$$

the substitution of the above series for w produces

$$f'(\xi) = a\Big\{1 - \tfrac{1}{2}\frac{bb}{aa}yy + \Big(\tfrac{1}{24}\frac{bd}{aa} + \tfrac{1}{8}\frac{bbc}{a^3} - \tfrac{1}{8}\frac{b^4}{a^4}\Big)y^4 - \Big(\tfrac{1}{720}\frac{bf}{aa} + \tfrac{1}{48}\frac{bcd}{a^3} - \tfrac{1}{12}\frac{b^3c}{a^4} + \tfrac{1}{16}\frac{b^5}{a^5}\Big)y^6 \ldots\Big\}.$$

Per page 144 of the Nachlass,

$$f(\xi) \;=\; f(x) - \tfrac{1}{2}byy + \tfrac{1}{24}dy^4 \ldots, \qquad \lambda \;=\; ay - \tfrac{1}{6}cy^3 + \tfrac{1}{120}ey^5 \ldots;$$

hence

$$\frac{\partial f(\xi)}{\partial x} = a\Big(1 - \tfrac{1}{2}\frac{c}{a}yy + \tfrac{1}{24}\frac{e}{a}y^4 \ldots\Big) = u, \qquad \frac{\partial \lambda}{\partial x} = a\Big(\frac{b}{a}y - \tfrac{1}{6}\frac{d}{a}y^3 + \tfrac{1}{120}\frac{f}{a}y^5 \ldots\Big) = v,$$

and consequently

$$\begin{aligned}
\frac{u}{f'(\xi)} &= \frac{\cos c}{n} = 1 + \Big(-\tfrac{1}{2}\frac{c}{a} + \tfrac{1}{2}\frac{bb}{aa}\Big)yy + \Big(\tfrac{1}{24}\frac{e}{a} - \tfrac{1}{24}\frac{bd}{aa} - \tfrac{1}{8}\frac{bbc}{a^3} + \tfrac{1}{8}\frac{b^4}{a^4}\Big)y^4 \ldots, \\
\frac{v}{f'(\xi)} &= \frac{\sin c}{n} = \frac{b}{a}y - \Big(\tfrac{1}{6}\frac{d}{a} - \tfrac{1}{2}\frac{b^3}{a^3}\Big)y^3 \ldots,
\end{aligned}$$

where c denotes again the Meridianconvergenz. Squaring and adding gives

$$\frac{1}{nn} = 1 + \Big(-\frac{c}{a} + 2\frac{bb}{aa}\Big)yy + \Big(\tfrac{1}{12}\frac{e}{a} - \tfrac{1}{12}\frac{bd}{aa} + \tfrac{1}{4}\frac{cc}{aa} - \tfrac{1}{2}\frac{bbc}{a^3} + 2\frac{b^4}{a^4}\Big)y^4 \ldots,$$

and therefore

$$2\log n = \Big(\frac{c}{a} - 2\frac{bb}{aa}\Big)yy + \Big(-\tfrac{1}{12}\frac{e}{a} + \tfrac{1}{12}\frac{bd}{aa} + \tfrac{1}{4}\frac{cc}{aa} - \tfrac{1}{2}\frac{bbc}{a^3}\Big)y^4 \ldots$$

[Der Logarithmus ist hier, wie in den nächsten Artikeln, der hyperbolische.]

[7.] Eine zweite Form für $\log n$ (scan p. 153)

Editorial gloss in source: `[Es ist auch]`; `[mithin]`; `[Nun ist]`.

A second route to $\log n$ uses the holomorphy of f directly. Since $f(x+iy)$ is analytic in $x+iy$ and $\xi + i\lambda$ is its conformal image, one has

$$\log n - ic = \log f'(\xi) - \log f'(x+iy);$$

taking real parts gives

$$\log n = \log f'(\xi) - \operatorname{Pars\,Real.} \log f'(x+iy).$$

Now

$$\log f'(\xi) = \log a - \frac{bb}{2aa}yy + \frac{aabd + 3abbc - 6b^4}{24a^4}y^4 \ldots,$$

and

$$\begin{aligned}
f'(x+iy) &= a\Big(1 + i\frac{b}{a}y - \tfrac{1}{2}\frac{c}{a}yy - \tfrac{1}{6}i\frac{d}{a}y^3 + \tfrac{1}{24}\frac{e}{a}y^4 + \cdots\Big), \\
\log f'(x+iy) &= \log a + \tfrac{1}{2}\Big(\frac{bb}{aa} - \frac{c}{a}\Big)yy + \Big(\frac{ae - 4bd + 3cc}{24aa} - \frac{(bb-ac)^2}{4a^4}\Big)y^4 \ldots \\
&\quad + i\Big(\frac{b}{a}y - \Big(\tfrac{1}{6}\frac{d}{a} - \tfrac{1}{2}\frac{bc}{aa} + \tfrac{1}{3}\frac{b^3}{a^3}\Big)y^3 \ldots\Big).
\end{aligned}$$

Hence

$$\text{Pars Real.} \log f'(x+iy) = \log a + \frac{bb-ac}{2aa}yy + \frac{a^2e - 4aabd - 3aacc + 12abbc - 6b^4}{24a^4}y^4 \ldots,$$

and therefore

$$\log n = \frac{ac-2bb}{2aa}yy - \frac{aae - 5abd - 3acc + 9bbc}{24a^3}y^4 \ldots \quad \text{[cf. S. 153];}$$

the meridian convergence appears in the imaginary part:

$$c = \frac{b}{a}y - \frac{aad - 3abc + 2b^3}{6a^3}y^3 \ldots \quad \text{[cf. S. 147].}$$

Re-inserting the values $a = f'(x)$, $b = f''(x)$, … from page 144 of the scan produces the closed expression

$$\log n = \frac{(1-ee\sin\varphi^2)^2}{2aa(1-ee)}yy - \frac{(1-ee\sin\varphi^2)^3}{12a^4(1-ee)^2}\left\{1 - 3ee + (ee + 15e^4)\sin\varphi^2 - 14e^4\sin\varphi^4\right\}y^4 \ldots,$$

[worin nun wieder a die halbe grosse Axe und e die Excentricität bedeutet].

[8.] Eine dritte Form: $\log n$ durch $\theta(x)$ und ihre Ableitungen (scan p. 154)

Editorial gloss in source: `[Oder setzt man:]`; `[so wird]`.

Set

$$\frac{1}{f'(x)} = \theta(x) = \alpha = \frac{1}{a},$$

and similarly

$$\theta'(x) = \beta = -\frac{b}{aa} = -\sin\varphi,$$

$$\theta''(x) = \gamma = \frac{2bb-ac}{a^3},$$

$$\theta'''(x) = \delta = \frac{-6b^3 + 6abc - aad}{a^4},$$

$$\theta^{\mathrm{IV}}(x) = \varepsilon = \frac{24b^4 - 36abbc + 6aacc + 8aabd - a^2e}{a^5}, \quad \text{u. s. w.,}$$

With these definitions the previous series collapses to

$$\log n = -\frac{\gamma}{2\alpha}yy + \frac{\alpha\alpha\varepsilon - 8\alpha\beta\delta - 8\alpha\gamma\gamma + 3\beta\beta\gamma}{24\alpha^3}y^4 \ldots,$$

$$n = 1 - \frac{\gamma}{2\alpha}yy + \frac{\alpha\alpha\varepsilon - 8\alpha\beta\delta + 3\beta\beta\gamma}{24\alpha^3}y^4 \ldots.$$

[9.] Die Laplace-artige Identität für $\log n$ (scan pp. 154–155)

This is the section the source TeX reduced to a five-line stub (`gauss_b01a_ch4.tex` lines 512–526). It is reconstructed in full from scan pp. 154–155.

One also has, on setting $\dfrac{1}{f'(\xi)} = \theta(\xi)$,

$$\frac{\partial^2 \log n}{\partial x^2} + \frac{\partial^2 \log n}{\partial y^2} = -\frac{\theta''(\xi)}{n\,\theta(\xi)}.$$

Suppose now

$$\frac{\theta''(\xi)}{\theta(\xi)} = -A - Byy - Cy^4 - \dots,$$

where $A, B, C, \dots$ depend only on ξ. Then integration in y (carrying x constant) yields

$$\log n = \tfrac{1}{2}Ayy + \tfrac{1}{12}By^4 + \tfrac{1}{30}Cy^6 + \cdots.$$

Differentiating the Ansatz once with respect to x produces $\frac{\partial A}{\partial x} = A'$, $\frac{\partial A'}{\partial x} = A''$, etc., and re-collecting terms gives

$$\begin{aligned}\log n = \tfrac{1}{2}Ayy + \tfrac{1}{12}\big(B - AA - \tfrac{1}{2}A''\big)y^4 \\ + \Big(\tfrac{1}{30}C - \tfrac{1}{180}B'' - \tfrac{1}{60}AB + \tfrac{1}{15}A^3 + \tfrac{1}{180}A'A' + \tfrac{1}{180}AA'' + \tfrac{1}{720}A^{\mathrm{IV}}\Big)y^6 \dots,\end{aligned}$$

which is the fully developed series; for $y = 0$ it reduces correctly to $\log n = 0$.

[10.] Die Differentialgleichung für $\log n$ direkt (scan p. 155)

Editorial gloss in source: `[Es ist]`.

Writing $\log n = N$, the Laplace-type identity above can be rearranged into the inhomogeneous form

$$\frac{\partial^2 N}{\partial x^2} + \frac{\partial^2 N}{\partial y^2} = \frac{(1 - ee\sin\Phi^2)^2}{aa\,nn\,(1 - ee)},$$

or, equivalently, by going back from $N = \log n$ to n itself,

$$\frac{n\,\partial^2 n}{\partial x^2} + \frac{n\,\partial^2 n}{\partial y^2} - \Big(\frac{\partial n}{\partial x}\Big)^2 - \Big(\frac{\partial n}{\partial y}\Big)^2 = \frac{(1 - ee\sin\Phi^2)^2}{aa\,(1 - ee)}.$$

This is the equation that the editorial bracket on p. 155 (`[Es ist]`) merely signposted; the inversion to n is what the source TeX dropped.

[11.] Umkehrung: x, y als Funktionen von ξ, λ (scan p. 155)

Editorial gloss in source: `[Beziehungen zwischen` x, y `und` ξ, λ`.]; [Die Umkehrung der Reihen für` ξ `und` λ `in Art. 6 ergibt, da nach Art. 8...]`.

Inversion of the series for ξ and λ developed in [6.], keeping in mind that by sect. [8.]

$$a = \frac{1}{\alpha}, \quad b = -\frac{\beta}{\alpha\alpha}, \quad c = -\frac{\gamma}{\alpha\alpha} + \frac{2\beta\beta}{\alpha^3}, \quad d = -\frac{\delta}{\alpha\alpha} + \frac{6\beta\gamma}{\alpha^3} - \frac{6\beta^3}{\alpha^4}, \quad \text{u. s. w.},$$

gives, after collecting up to fourth order in λ:

$$\begin{aligned} x &= \xi - \tfrac{1}{2}\alpha\beta\lambda\lambda + \tfrac{1}{24}(\alpha^3\delta - 2\alpha\alpha\beta\gamma - 5\alpha\beta^3)\lambda^4 \ldots, \\ y &= \alpha\lambda - \tfrac{1}{6}(\alpha\alpha\gamma - 2\alpha\beta\beta)\lambda^3 \ldots. \end{aligned}$$

[12.] Eine implizite Definition $\theta(t) = e^{\vartheta(t)}$ (scan p. 155)

Editorial gloss in source: `[also nach Art. 8:` $\theta(t) = e^{\vartheta(t)}$`]`.

For the further developments it is convenient to put

$$\int e^{-\vartheta(t)}\, dt = f(t), \qquad \text{i. e., following sect. [8.], } \theta(t) = e^{\vartheta(t)}.$$

Writing the successive derivatives of ϑ as $\vartheta'(x) = a_1$, $\vartheta''(x) = \beta_1$, $\vartheta'''(x) = \gamma_1$, $\vartheta^{\mathrm{IV}}(x) = \delta_1$, $\vartheta^{\mathrm{V}}(x) = \varepsilon_1$, one obtains successively

$$\begin{aligned} f''(x) &= -a_1 f'(x), \\ f'''(x) &= (a_1a_1 - \beta_1)f'(x), \\ f^{\mathrm{IV}}(x) &= (-a_1^3 + 3a_1\beta_1 - \gamma_1)f'(x), \\ f^{\mathrm{V}}(x) &= (a_1^4 - 6a_1a_1\beta_1 + 4a_1\gamma_1 + 3\beta_1\beta_1 - \delta_1)f'(x), \\ f^{\mathrm{VI}}(x) &= (-a_1^5 + 10a_1^3\beta_1 - 10a_1a_1\gamma_1 - 15a_1\beta_1\beta_1 + 5a_1\delta_1 + 10\beta_1\gamma_1 - \varepsilon_1)f'(x), \end{aligned}$$

the rule being clear from the pattern: each subsequent $f^{(n+1)}(x)/f'(x)$ is the previous one differentiated and then expressed in $\vartheta', \vartheta'', \ldots$. By substituting these into the series of [6.], one obtains the corresponding inversions for ξ and λ,

$$\begin{aligned} \xi = x + \tfrac{1}{2}a_1yy + (\tfrac{1}{6}a_1^3 + \tfrac{1}{8}\beta_1)y^4 & \\ + \big(\tfrac{1}{24}a_1^5 + \tfrac{1}{12}a_1a_1\beta_1 + \tfrac{1}{24}a_1\gamma_1 - \tfrac{1}{8}\beta_1\beta_1 + \tfrac{1}{48}\delta_1\big)y^6 \ldots, & \end{aligned}$$

$$\lambda = y\, f'(x)\Big\{1 - \tfrac{1}{6}(a_1a_1 - \beta_1)yy + \tfrac{1}{120}(a_1^4 - 6a_1a_1\beta_1 + 4a_1\gamma_1 + 3\beta_1\beta_1 - \delta_1)y^4 \ldots\Big\}.$$

(The first inverse series is the analogue of $\xi = x - w + \cdots$ in [6.]; the second is the imaginary part.)

[13.] Berechnung der ebenen rechtwinkligen Coordinaten aus der geographischen Breite und Länge (scan p. 156)

Editorial gloss in source: `[Berechnung der ebenen rechtwinkligen Coordinaten aus der geographischen Breite und Länge.]`.

For the reciprocal task (given the geographic latitude Φ and longitude λ, compute the plane rectangular coordinates x, y), denote by $\mathfrak{g}$ the inverse function of f, so that $\mathfrak{g}(F(\Phi)) = x$, and set $\mathfrak{g}(F) = \mathfrak{F}$. One has then

$$\begin{aligned} x + iy &= \mathfrak{g}(F + i\lambda), \\ x &= \mathfrak{g}(F) - \tfrac{1}{2}\lambda^2\mathfrak{g}''(F) + \tfrac{1}{24}\lambda^4\mathfrak{g}^{\mathrm{IV}}(F)\ldots, \\ y &= \lambda\,\mathfrak{g}'(F) - \tfrac{1}{6}\lambda^3\mathfrak{g}'''(F) + \tfrac{1}{120}\lambda^5\mathfrak{g}^{\mathrm{V}}(F)\ldots, \\ \mathfrak{g}(F) &= X, \end{aligned}$$

where X is the abscissa belonging to the intersection of the parallel of latitude Φ with the main meridian (so that $f(X) = \mathfrak{g}(F)\,\big|_{F=F(\Phi)} = X$ by sect. [1.]); hence

$$\frac{dX}{d\Phi} = \frac{1}{F'(\Phi)}.$$

Therefore

$$\mathfrak{g}'(F) = \frac{1}{F'(\Phi)} = \theta(X) = h, \qquad c = \cos\Phi, \qquad s = \sin\Phi, \qquad p = \sqrt{(1 - ee\sin\Phi^2)},$$

where, as before, a denotes the semimajor axis and e the eccentricity. Following the rule of differentiation given on page 144 of the scan,

$$\frac{dh}{d\Phi} = c + 2c^3, \qquad \frac{dh}{d\Phi} = \frac{ee}{1-ee} - h,$$

or, dropping the higher powers of ee,

$$\frac{dh}{d\Phi} = c + ee\,c^3;$$

and continuing,

$$\frac{ds}{d\Phi} = \frac{pp(1-ee)}{1-ee} = cc + 3c^3, \quad \frac{dc}{d\Phi} = -\frac{ppc}{1-ee} = -s(c + 2c^3), \quad \frac{dh}{d\Phi} = -sh.$$

If now one sets $\mathfrak{g}^{n+1}(F) = hu$, with u a rational function of c, then

$$[\mathfrak{g}^{n+1}(F)] = -sh\left[u\,|n - 2cc - 3c^3| + \frac{du}{dc}(c + 3c^3)\right],$$

and (neglecting higher powers of ee)

$$\mathfrak{g}^{n+1}(F) = -h\left[u(1 - 2cc - 3c^4) + \frac{du}{dc}(1-cc)(c + ee\,c^3)\right].$$

By the same rule, but with u a rational function of c, one obtains the table (cf. p. 157 of the scan):

$$\begin{aligned}
[\mathfrak{g}'(F)] &= h,\\
\mathfrak{g}''(F) &= -hs,\\
\mathfrak{g}'''(F) &= +h(1-2cc-3c^4),\\
\mathfrak{g}^{\mathrm{IV}}(F) &= -hs\,(1-6cc-9c^4-43c^6),\\
\mathfrak{g}^{\mathrm{V}}(F) &= +h\big(1-20cc+(24-64\,ee)c^4+(77\,ee-24\,ee^2)c^6+28\,ee^2c^8\big),
\end{aligned}$$

etc.

Consequently, if x is measured as on the scan, with X taken positive toward south and y having the same sign as λ:

$$\begin{aligned}
x &= X - \frac{ee}{8\sqrt{1-ee\,ee}}\lambda\lambda + \frac{ee}{384(1-ee\,ee)}\lambda^4\dots,\\
y &= \frac{ae}{\sqrt{1-ee\,ee}}\lambda - \frac{ae}{6\sqrt{1-ee\,ee}}(1-2ee-3c^4)\lambda^3\\
&\qquad + \frac{ae}{120\sqrt{1-ee\,ee}}(1-20cc+(24-58\,ee)c^4+\dots)\lambda^5\dots
\end{aligned}$$

which is the development for the plane rectangular coordinates from Φ and λ. The coefficients are exactly as on scan p. 157 (third paragraph).

Cross-reference: closing glosses of `ch3.tex` (Conforme Abbildung des Sphäroids in der Ebene, scan pp. 147–148)

The two bracketed glosses in `gauss_b01a_ch3.tex` around lines 730–755 (`[oder genähert]`, `[Ferner hat man:]`, `[oder, wenn` $\frac{f''(x)}{f'(x)} = h(x)$ `gesetzt wird,` $c = yh(x) - \frac{1}{3}y^3h''(x)\dots$`])` are likewise five-line stubs of multi-line developments; for completeness their fully-typeset content is included here so that the file self-contains the entire chapter on *Conforme Abbildung des Sphäroids in der Ebene.*

Closing of [2.] (scan p. 147)

From the equation that represents the meridian of longitude λ in the plane,

$$\lambda = \text{const} = yf'(x) - \tfrac{1}{6}y^3f'''(x)\dots,$$

one obtains

$$\frac{dy}{dx} = -\frac{yf''(x) - \frac{1}{2}y^3f^{\mathrm{IV}}(x)\dots}{f'(x) - \frac{1}{2}yyf'''(x)\dots},$$

and therefore

$$\operatorname{tang} c = \frac{yf''(x) - \frac{1}{2}y^3f^{\mathrm{IV}}(x)\dots}{f'(x) - \frac{1}{2}yyf'''(x)\dots} = y\frac{f''(x)}{f'(x)}\Big\{1 - yy\Big(\tfrac{1}{6}\frac{f^{\mathrm{IV}}(x)}{f''(x)} - \tfrac{1}{2}\frac{f'''(x)}{f'(x)}\Big)\dots\Big\}.$$

Therefore, by p. 144,

$$\operatorname{tang} c = D \cdot \frac{\sqrt{(1-ee\sin\varphi^2)}}{a}\, y \operatorname{tang}\varphi,$$

where

$$\begin{aligned}\log\operatorname{hyp} D &= \Big(-\tfrac{1}{6}\frac{f^{\mathrm{IV}}(x)}{f''(x)} + \tfrac{1}{2}\frac{f'''(x)}{f'(x)}\Big) yy \ldots\\ &= -\tfrac{1}{2} yy\, \frac{(1-ee\sin\varphi^2)^3}{aa(1-ee)^3}\,(1-3ee+2ee\sin\varphi^2)\ldots\end{aligned}$$

or, in the approximation,

$$= -\tfrac{1}{2}\frac{yy}{aa}(1-ee).$$

Furthermore,

$$c = y\frac{f''(x)}{f'(x)} - \tfrac{1}{3}y^3\Big\{\frac{f^{\mathrm{IV}}(x)}{f'(x)} - \frac{3f''(x)f'''(x)}{(f'(x))^2} + 2\Big(\frac{f''(x)}{f'(x)}\Big)^3\Big\}\ldots$$

or, abbreviating $f''(x)/f'(x) = h(x)$,

$$c = y\,h(x) - \tfrac{1}{3}y^3\,h''(x)\ldots.$$

The associated rational functions in φ are

$$\begin{aligned}h(x) &= \frac{(1-ee\sin\varphi^2)^{3/2}\sin\varphi}{a\cos\varphi},\\ h'(x) &= \frac{1-ee\sin\varphi^2}{aa(1-ee)\cos\varphi^2}\big\{1-2ee\sin\varphi^2+ee\sin\varphi^4\big\},\\ h''(x) &= \frac{2(1-ee\sin\varphi^2)^{3/2}\sin\varphi}{a^3(1-ee)^2\cos\varphi^4}\big\{1-3ee+(2ee+4e^4)\sin\varphi^2-(ee+5e^4)\sin\varphi^4+2e^4\sin\varphi^6\big\}\\ &= \frac{2(1-ee\sin\varphi^2)^{3/2}\sin\varphi}{a^3(1-ee)^2\cos\varphi^4}\big\{(1-ee)^2-ee(1-ee)\cos\varphi^4-2e^4\cos\varphi^6\big\}.\end{aligned}$$

Therefore

$$\begin{aligned}c = \frac{\sqrt{(1-ee\sin\varphi^2)}}{a}\operatorname{tang}\varphi\cdot y - \tfrac{1}{3}\frac{(1-ee\sin\varphi^2)^{3/2}\sin\varphi}{a^3(1-ee)^2\cos\varphi^3}\big\{1-3ee+(2ee+4e^4)\sin\varphi^2\\ -(ee+5e^4)\sin\varphi^4+2e^4\sin\varphi^6\big\}y^3\ldots\end{aligned}$$

or, in the alternative form,

$$\begin{aligned}c = &\frac{\sqrt{(1-ee\sin\varphi^2)}}{a}\operatorname{tang}\varphi\cdot y\\ &-\tfrac{1}{3}\frac{(1-ee\sin\varphi^2)^{3/2}\sin\varphi}{a^3(1-ee)^2\cos\varphi^3}\big\{(1-ee)^2-ee(1-ee)\cos\varphi^4-2e^4\cos\varphi^6\big\}y^3\ldots\end{aligned}$$

Introducing λ as the principal independent variable one also finds

$$c = \lambda\sin\varphi - \frac{(1-ee\sin\varphi^2)^{3/2}\operatorname{tang}\varphi}{6a^3(1-ee)^3}\big\{(1-ee)^2-3ee(1-ee)\cos\varphi^2-4e^4\cos\varphi^4\big\}y^3\ldots.$$

[3.] (scan p. 148, glosses `[wo` $H = \frac{1}{2}k\rho^2 \cdot 10^7$, $\log H = 5{,}531\,8128 - 10]$ expanded)

For numerical computation the most convenient formulae are obtained by setting

$$L = \frac{(1 - ee\sin\varphi^2)^{3/2}\, y}{a\rho\cos\varphi},$$
$$M = \frac{(1 - ee\sin\varphi^2)^2\ \operatorname{tang}\varphi \cdot yy}{2aa(1-ee)\rho},$$
$$N = \frac{(1 - ee\sin\varphi^2)^{3/2}}{a\rho}\ \operatorname{tang}\varphi \cdot y.$$

Then

$$\log\lambda = \log L - A\cdot LL,$$
$$\log(\varphi - \Phi) = \log M - B\cdot LL,$$
$$\log c = \log N - C\cdot LL,$$

where the logarithms of A, B, C are most conveniently taken from a separate table with argument φ. For computing this table use

$$A = H\Big\{0{,}75 - \tfrac{1}{4}\cos 2\varphi + \tfrac{ee}{2(1-ee)}\cos\varphi^4\Big\},$$
$$B = H\Big\{0{,}75 + \tfrac{2-11ee}{4(1-ee)}\cos\varphi^2 + \tfrac{5ee}{2(1-ee)}\cos\varphi^4 - \tfrac{e^4}{(1-ee)^2}\cos\varphi^6\Big\},$$
$$C = H\Big\{1 - \tfrac{ee}{1-ee}\cos\varphi^4 - \tfrac{2e^4}{(1-ee)^2}\cos\varphi^6\Big\},$$

where (the gloss explicitly):

$$H = \tfrac{1}{2}k\rho\rho\cdot 10^7, \qquad \log H = 5{,}531\,8128 - 10,$$

with k the modulus of Briggsian (common) logarithms and $\rho = 1/206264{,}8\ldots$.

The corrections $A\cdot LL$, $B\cdot LL$, $C\cdot LL$ are obtained in units of the seventh decimal; λ, $\varphi - \Phi$ and c come out in seconds. The subsequent table of $\log A$, $\log B$, $\log C$ in the source TeX is based on the flattening 1/302,68 (the Hannover value); cf. scan p. 149.

Audit cross-references

- `gauss_b01a_ch4.tex` lines 480–530 ([6.]–[9.]): expanded above (Sects. [6.]–[9.] of this file).
- Source TeX line 512–517 (the explicit 5-line stub of [9.]): now Section [9.] of this file. The whole Laplace-type identity plus its integration to a series in y is restored.
- Source TeX line 397 ([Tabular data for $\log A$, $\log B$, $\log C$ follows — transcribed as placeholder]): the table itself is in the Hannover scan p. 149 and is referenced but not retyped here (the audit explicitly noted this as a known gap for Phase 2 manual table re-entry).

- Source TeX line 487–491 ([6.]–[7.] "Weitere Entwicklungen für die Coordinatenberechnung"): expanded above to scanned content.
- Source TeX line 493–510 ([8.]–[9.] "Allgemeine Formeln für die Differentiale"): fully written out as Sect. [8.] (definitions of $\alpha, \beta, \gamma, \delta, \varepsilon$) and Sect. [9.] of this file (the Laplace-type identity).
- Source TeX line 529–536 ([10.]–[14.] "Entwicklung der Grundformeln ..." — summarised in source as [Die Entwicklung beruht auf folgender Aufgabe.]): the auxiliary lemma is in the source TeX correctly; the surrounding [Schlussbemerkungen und Tabellen] stub is intentionally left as-is (numerical tables only).

Expanded against scan, end of file.